Exact spike train inference via ℓ_0 optimization

- Annals of Applied Statistics (2018)
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Why analyze spike trains?

Contributions

Model

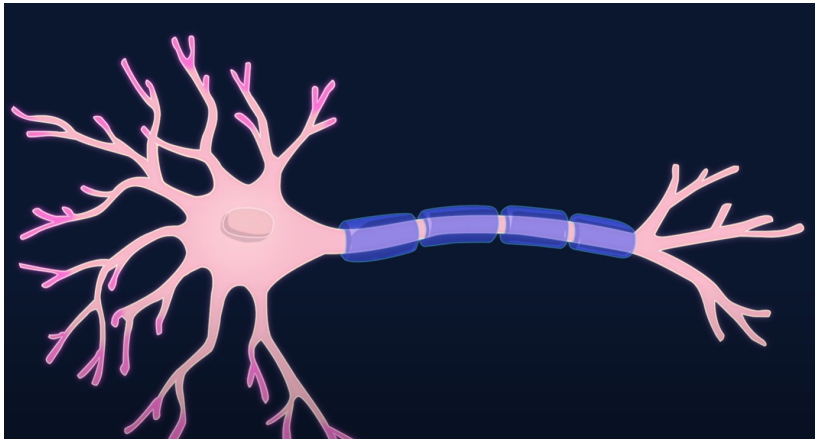
Optimization problem

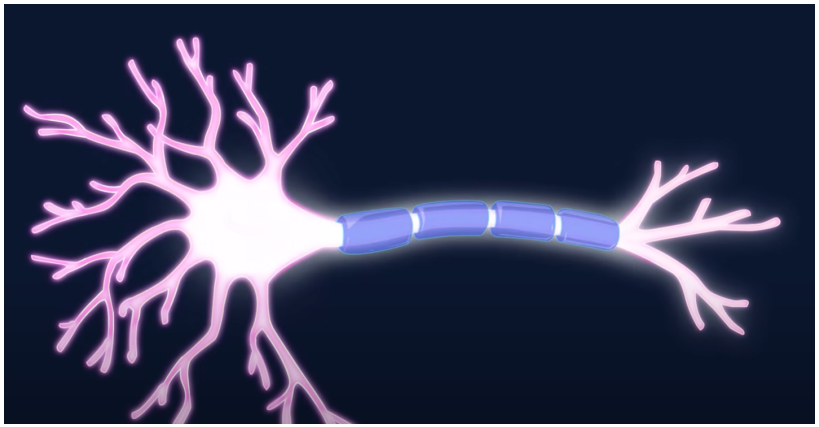
$O(T^2)$ algorithm

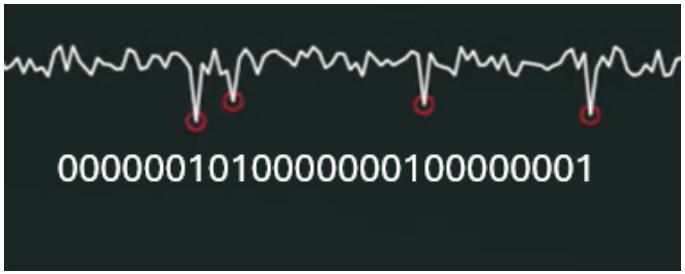
Conclusion

- Neural spike train
- Why do we analyze them

Spike train inference

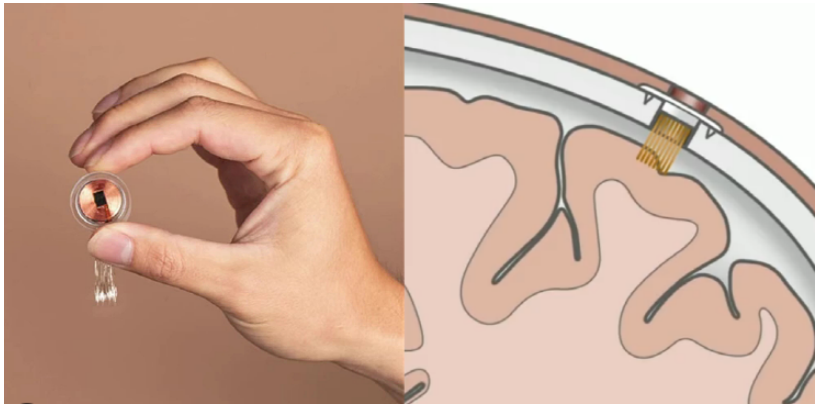






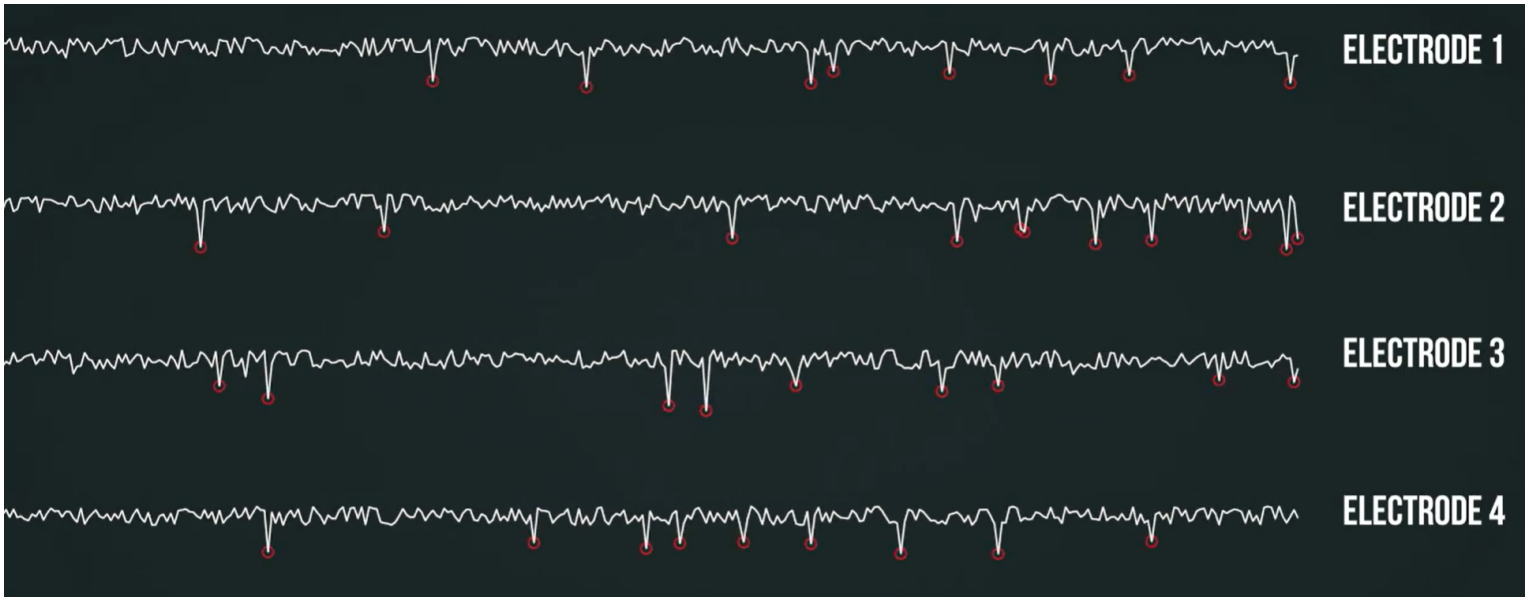
- Brain operates through electrical events known as *action potentials*
- Information is primarily encoded as sequences of these action potentials: binary patterns (01010...) known as *spike trains*
- *Spike train inference*: recovering these spike sequences from recorded neural electrical signals

Connection to ML



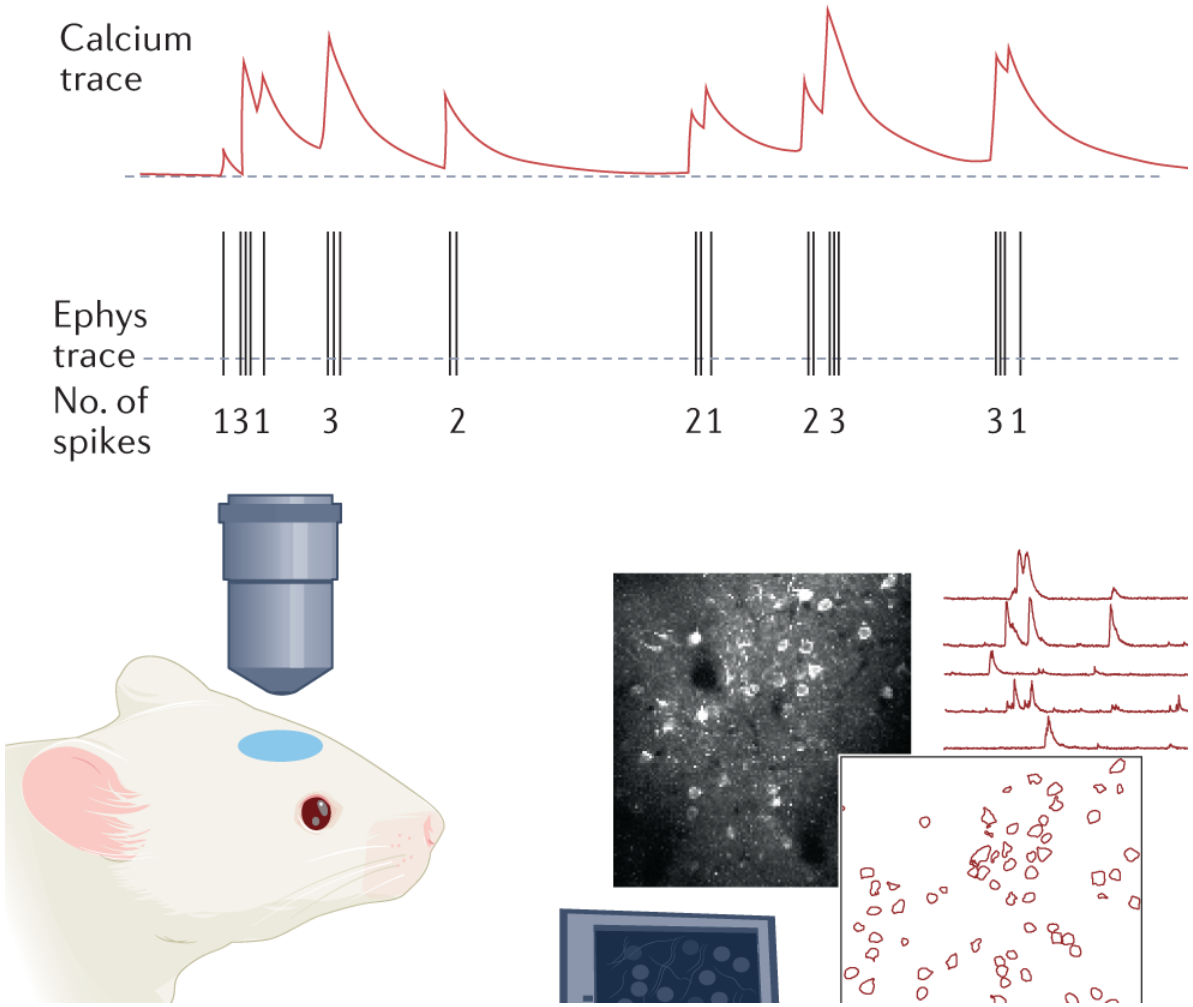
- Spike train inference empowers machine learning tasks such as both *brain decoding* (inferring thoughts from neural activity) and *brain encoding* (translating intent into neural signals for action)

Electrodes



- Spike train inference is relatively straightforward when high-quality electrical signals are directly recorded (e.g., via simple thresholding methods)
- However, such recordings are often invasive and do not scale well to monitoring large populations of neurons simultaneously

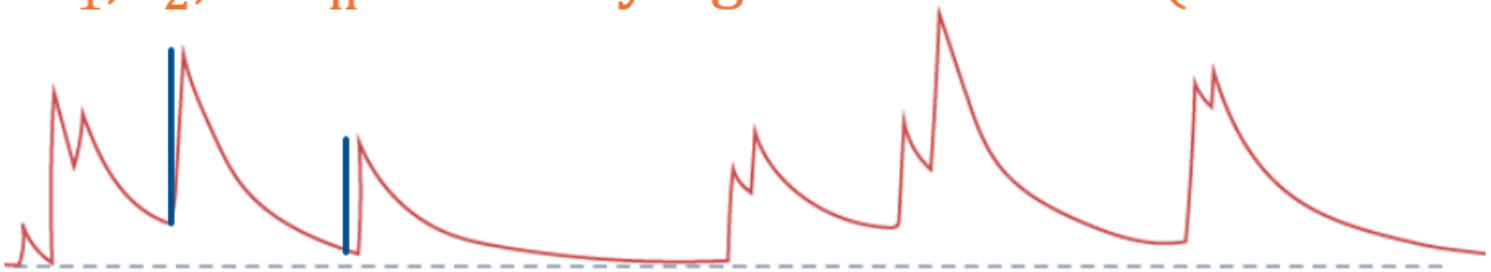
Calcium imaging: slow but scalable



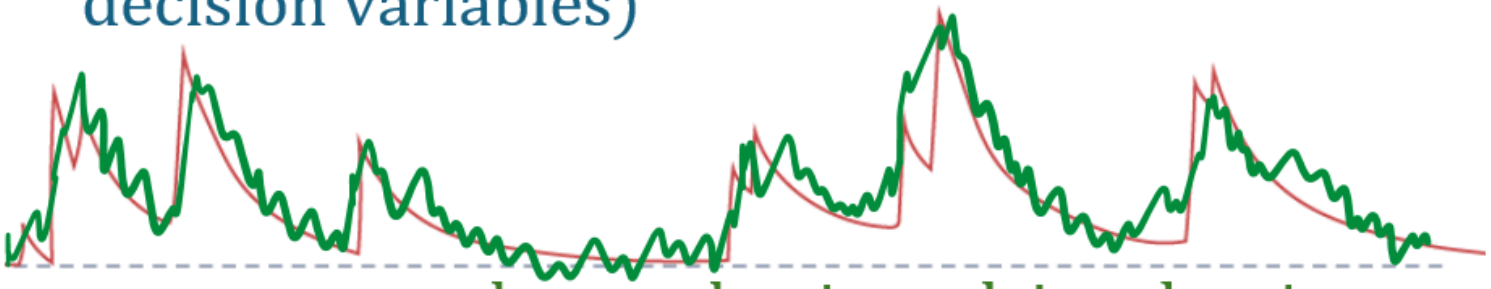
- Calcium imaging enables recording from thousands of neurons simultaneously, but its slow dynamics cause signals to overlap
- new spikes can occur before previous responses have fully decayed.

Mixed Integer Programming Formulation

$s_1, s_2, \ldots s_n$: underlying calcium level (decision variables)



$x_1, x_2, \ldots x_n$: effect of action potential (0 if no spike; decision variables)



$r_1, r_2, \ldots r_n$: observed noisy calcium level

To filter out the noise, we solve least squares:

$$\min_{\mathbf{s}} \frac{1}{2} \sum_{i=1}^{n+1} (s_i - r_i)^2$$

Mixed Integer Programming Formulation

- AR(1) model:

$$s_{i+1} = \alpha s_i + x_i$$

  - Calcium level always decays by $\alpha \in (0, 1)$
  - If $x_i > 0$, there is an effect of an action potential

- Sparse spikes: add an on-off switch $z_1, \ldots, z_n$ and link it to x_i:

$$\begin{aligned}
\min_{\mathbf{s},\mathbf{x},\mathbf{z}} \quad & \frac{1}{2}\sum_{i=1}^{n+1}(s_i - r_i)^2 + \lambda \sum_{i=1}^{n} z_i \\
\text{s.t.} \quad & x_i(1 - z_i) = 0, \quad z_i \in \{0, 1\}, \quad i = 1, \ldots, n \\
& x_i = s_{i+1} - \alpha s_i, \quad i = 1, \ldots, n \\
& s_1 = 0
\end{aligned}$$

Contributions

- Formulates the spike detection as ℓ_0 regularized optimization
- Transforms the problem in to change-point detection problem
- Devise an $O(T^2)$ solver based on dynamic programming
- Faster and more accurate than ℓ_1 regularization.

Three stages of models

- Spike sequence (fixed, unobserved)
- Underlying calcium concentration process (fixed, unobserved)
- Noise-added observations (random, observed)

1. Spike sequence

$s_t \geq 0,\ t1, \ldots, T$

- Example: 0, 3.5, 0, 0, 1.1, 0, 0, 2, ...
- If $s_t > 0$: spike at time t
- If $s_t = 0$: no spike at time t
- Fixed unknown parameter of interest
- Sparsity assumption

2. Underlying calcium concentration

- First-order autoregressive process

$$C_t = \gamma C_{t-1} + s_t, \quad t = 2, 3, \ldots, T$$

- Spike appears abrupt, and gradually disappears.
- The effect of spikes can overlap
- Fixed unknown parameter

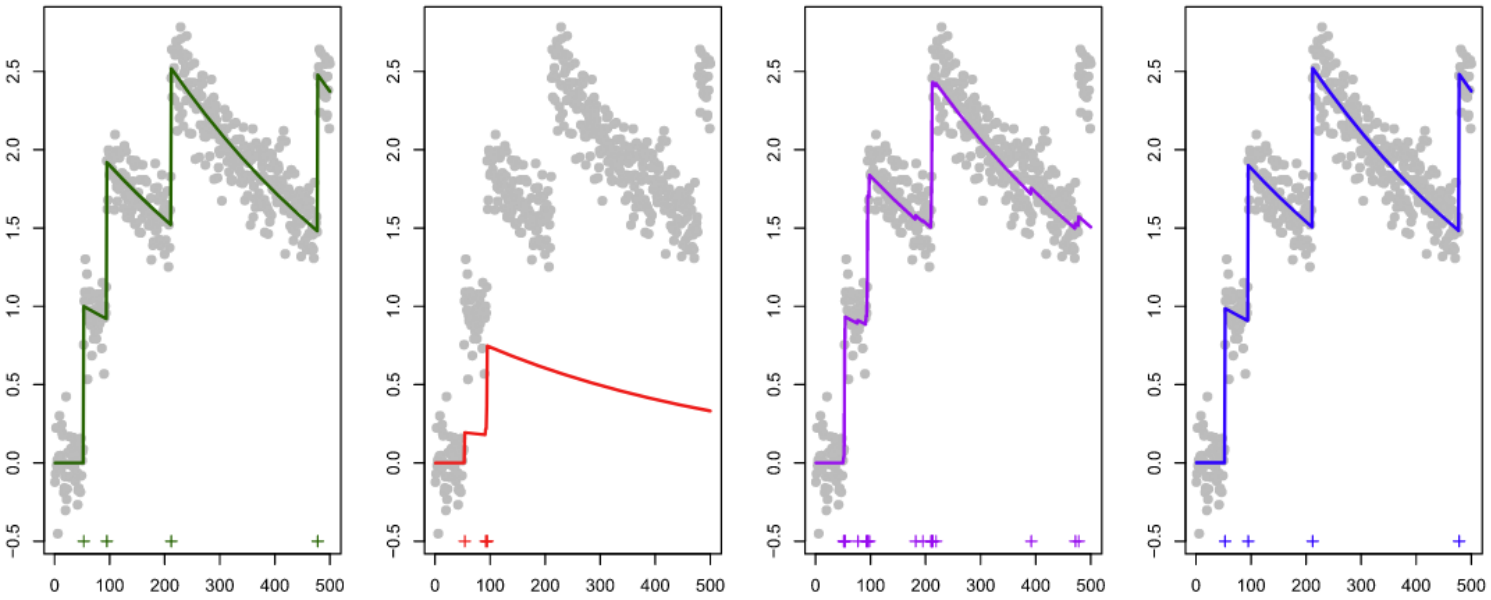


Figure: Spike appears abrupt, and gradually disappears

3. Observed fluorescence process

$$Y_t = c_t + \varepsilon_t, \quad \varepsilon_t \sim_{\text{ind}} (0, \sigma^2)$$

for $\quad t = 1, \ldots, T$

Optimization problem

$$\min_{c_1,\ldots,c_T,\, s_2,\ldots,s_T} \frac{1}{2}\sum_{t=1}^{T}(y_t - c_t)^2 + \lambda \sum_{t=2}^{T} \mathbb{1}(s_t = 0)$$
$$\text{s.t. } s_t = c_t - \gamma c_{t-1}.$$

- Regularization $\lambda \mathbb{1}(c_t \neq \gamma c_{t-1})$ ties c_t and c_{t+1}.
- This tie is active only when $c_t = \gamma C_{t-1}$ $(s_t = 0)$ (no spike)

Another viewpoint

Thus, if the solution sets $s_t > 0$, than we actually solved the problem that decouples into $[1, t-1]$ and $[t+1, T]$, and t is a changepoint.

Thus, problem is equivalent to the changepoint detection problem:

$$\text{(spike magnitude} \longleftrightarrow \text{timepoint)}$$

$$\begin{array}{ccccccccc} 0 & 1.1 & 0 & 0 & 3.5 & 0 & 2.33 & \cdots \\ & 2 & & & 5 & & 8 & \end{array}$$

Proposition 1

The problem is equivalent to

$$\min_{0=\tau_0<\tau_1<\cdots<\tau_k<\tau_{k+1}=T} \left\{ \sum_{j=0}^{k} \mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda k \right\}.$$

for two timepoints $a = \tau_j + 1, \quad b = \tau_{j+1}$,

$$\mathcal{D}(a,b) := \arg \min_{c_a, c_{a+1}, \ldots, c_b} \frac{1}{2} \sum_{t=a}^{b} (y_t - c_t)^2$$
$$\text{s.t.} \quad c_t = \gamma c_{t-1}.$$

$$\mathcal{D}(a,b) := \arg \min_{c_a, c_{a+1}, \ldots, c_b} \frac{1}{2} \sum_{t=a}^{b} (y_t - c_t)^2$$
$$\text{s.t.} \quad c_t = \gamma c_{t-1}.$$

- Measures homogeneity of the measurement
- Has a closed-form formula (Proposition 2).

$$\min_{0=\tau_0<\tau_1<\cdots<\tau_k<\tau_{k+1}=T} \left\{ \sum_{j=0}^{k} \mathcal{D}(\tau_j + 1, \tau_{j+1}) + \lambda k \right\}.$$

Find not too many timepoints such that between two consecutive timepoints, the measurements are homogeneous

Apply the dynamic programming algorithm proposed by Jackson et al. (2005, *IEEE Signal Processing Letters*) developed for changepoint detection.

$$F(T) := \min_{0=\tau_0<\tau_1<\cdots<\tau_k<\tau_{k+1}=T} \left\{ \underbrace{\sum_{j=0}^{k} \mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda k}_{\text{(A)}} \right\} \quad (k \text{ intervals})$$

$$\text{(A)} = \sum_{j=0}^{k} [\mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda] - \lambda \quad (k+1 \text{ terms})$$

$$= \underbrace{\sum_{j=0}^{k-1} [\mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda] - \lambda}_{\tau_0, \ldots, \tau_k\text{: } k \text{ intervals, contributes to } F(\tau_k)} + \underbrace{\mathcal{D}(\tau_k+1, \tau_{k+1}) + \lambda}_{\tau_k+1,\, \tau_{k+1}}$$

Dynamic programming

Set $F(0) = -\lambda$

$$F(1) = \min_{\tau \in \{0\}} \{F(\tau) + \mathcal{D}(y_{\tau+1:1}) + \lambda\}$$

$$F(2) = \min_{\tau \in \{0,1\}} \{F(\tau) + \mathcal{D}(y_{\tau+1:2}) + \lambda\}$$

$\mathcal{D}$ calculation in constant time, at step t, reads t data:
$O(\sum_{t=1}^{T} T) = O(T^2)$

- Formulates the spike detection as ℓ_0 regularized optimization
- Transforms the problem in to change-point detection problem
- Devise an $O(T^2)$ solver based on dynamic programming
- Faster and more accurate than ℓ_1 regularization (numerically).
- No statistical theory, just optimization